

README.md (GitHub Repository)

Nairobi Nexus Automated Weekly Newsletter Generator

#Project Overview

This repository contains the production-grade implementation of an automated weekly newsletter generator for Nairobi Nexus. It fetches news from five Kenyan sources (TechCabal, Business Daily, The Star, The Standard, Kenyan Wallstreet), extracts headlines using dedicated AI agents, synthesises them into a witty newsletter intro (with Kenyan slang and Markdown), validates the output, logs full governance data, and emails the final draft. It also includes a global error workflow and an audit workflow that answers "What happened on any given day?" in plain English.

****Key features:****

- Parallel, fault-tolerant fetching (one source failure does not stop the rest)
- Truncation to prevent out-of-memory errors
- Backup AI agent (fallback) when the primary Chief Editor fails
- Quality Assurance (QA) agent that validates length, slang, Markdown, and relevance, and generates a weekly ****parting thought****
- Governance sheet logging every run: sources succeeded/failed, model used, fallback triggered, QA results, etc.
- Audit workflow (webhook + AI) to answer questions about past executions
- HTML email with parting thought

Repository Structure

```
|— workflows/
| |— main_workflow.json # Production newsletter workflow
| |— error_workflow.json # Global error handler
| |— audit_workflow.json # Audit answer workflow
| |— test_backup_ai.json # Test copy for backup AI validation
| |— test_error_workflow.json # Test copy for global error validation
|— docs/
|   — project_overview.pdf # Formal documentation
|— html/
|   — audit.html # Simple HTML form for managers (POST request)
|— README.md # This file
|— LICENSE # (optional)
```

Installation

Prerequisites

- n8n instance (cloud or self-hosted, version $\geq$ 2.21.8)
- Google Sheets account (for NewsletterManager spreadsheet)
- Gmail account (for sending emails)
- API keys for:
 - Groq (llama-3.3-70b-versatile, llama-3.1-8b-instant)
 - Mistral (magistral-small-latest, magistral-medium-latest, mistral-large-latest)
 - OpenAI (gpt-4o-mini – used for QA, audit, and error AI)
 - Google Gemini (optional, for some Field Reporters)
 - OpenRouter (optional)

Setup Steps

1. **Import workflows** into n8n:
 - Create three new workflows: `Main Workflow`, `Error Workflow`, `Audit Workflow`.
 - Copy the JSON content from the files in `/workflows/` into the respective workflows.
2. **Configure credentials**:
 - Open each workflow and re-authenticate:
 - Google Sheets (point to your `NewsletterManager` spreadsheet)
 - Gmail (OAuth2)
 - Each AI provider (Groq, Mistral, OpenAI, etc.)
3. **Create the Google Sheet (`NewsletterManager`)** with two tabs:
 - `Drafts` – columns: Date, Successful Sources, Draft Body, Model Used
 - `Governance` – columns: Timestamp, Sources Succeeded, Sources Failed, Model Used, Workflow Version, Output Status, Fallback Triggered, QA Passed, QA Issues
4. **Activate the workflows**:
 - Publish (activate) the main, error, and audit workflows.
 - Set the main workflow's trigger to **Schedule** (Friday 14:00) or keep the webhook for manual testing.
 - In the main workflow's Settings, set **Error Workflow** to the error workflow.
5. **Test**:
 - Call the main workflow's production webhook (or wait for the schedule) to generate a newsletter.
 - Check the email, the Drafts sheet, and the Governance sheet.
 - Force a fallback: temporarily change the primary Chief Editor's API key to an invalid one and re-run.

- Force a fatal error: insert a Code node with `throw new Error("test")` in a copy of the main workflow (or temporarily break an HTTP node) and verify the error workflow sends an alert.

Usage

For the Marketing Team (Weekly Newsletter)

- The workflow runs automatically every Friday at 2:00 PM (Africa/Nairobi).
- The team receives an email with the newsletter intro and parting thought.
- The `Drafts` tab stores the full history of generated newsletters.

For the Manager (Audit)

1. Open the `audit.html` file (provided in `/html/`) in a browser.
2. Select a date and click "Get Report".
3. An email with the answer will be sent to the configured address.

Alternatively, call the audit webhook with a POST request:

```
```bash
curl -X POST https://your-n8n-instance/webhook/audit \
 -H "Content-Type: application/json" \
 -d '{"date": "2026-06-01"}'
```

## For the Operations Team (Error Alerts)

- If the main workflow fails fatally, the error workflow sends an email with a hypothesis and recommended action.

## Environment Requirements

- n8n version 2.21.8 or higher
- Node.js (for self-hosted) – not required for n8n cloud
- Adequate memory (Pro plan recommended for heavy parallel processing) – truncation reduces memory pressure.

## Troubleshooting

Issue	Likely cause	Solution
Google Sheets node has no output	Filter condition not matching column name	Check column headers; use exact case.
AI Agent outputs "Model output doesn't fit required format"	Output parser schema mismatch	Inspect the AI's raw output and adjust schema or prompt.
Fallback not triggered	Primary node not set to <code>Continue (using error output)</code>	Change the On Error setting.
Out-of-memory	Large HTML pages	Truncation nodes already reduce size; if still occurs, reduce slice size to 3000.
Email empty	Expression path wrong	Use <code>\$node["Extract Governance &amp; Intro"].json.intro</code> directly.

## Deployment

1. Production – Activate the main workflow with the Schedule Trigger. Ensure error workflow is linked and active.
2. Backup – Keep the `test_backup_ai.json` and `test_error_workflow.json` for validation before any changes.

## License

This project is for educational purposes (Moringa School capstone). Not for commercial use without permission.

## Acknowledgments

- n8n team for the automation platform.
- AI providers (Groq, Mistral, OpenAI) for language models.
- Moringa School instructors for guidance.